

QFT (Ch 3 : 3.1)

$$\mathcal{H} \equiv \int d^3x \mathcal{H}$$

Similarly

$$L = \int d^3x \mathcal{L}$$

↑
Lag. density

defined by
Legendre
transform

$$\underline{L[\phi, \dot{\phi}] = \pi[\phi, \dot{\phi}] \dot{\phi} - \mathcal{H}[\phi, \pi[\phi, \dot{\phi}]]}$$

$$w/ \dot{\phi} = \partial_t \phi, \quad \pi[\phi, \dot{\phi}] \text{ def by } \frac{\partial \mathcal{H}[\phi, \pi]}{\partial \pi} = \dot{\phi}$$

Inverse Legendre transform implies

$$\mathcal{H}[\phi, \pi] = \pi \dot{\phi}[\phi, \pi] - \mathcal{L}[\phi, \dot{\phi}[\phi, \pi]]$$

$$\text{with } \wedge \text{ defined by } \frac{\partial \mathcal{L}[\phi, \dot{\phi}]}{\partial \dot{\phi}} = \pi$$

Eu. Lagrangian:

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi) (\partial_\mu \phi) - \mathcal{V}[\phi] = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\vec{\nabla} \phi)^2 - \mathcal{V}[\phi]$$

where $\mathcal{V}[\phi]$ is potential (density)

π is then $\dot{\phi}$ allowing us to find

$$\mathcal{H} = \frac{1}{2} \pi^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \mathcal{V}[\phi]$$

$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\nabla \phi)^2 + \mathcal{V}[\phi]$$

- Some $\mathcal{L}$ cannot produce closed form for ϕ & therefore very hard to work with

- Don't work with Minkowski $|t|$ density since they are components of P^μ & $T_{\mu\nu}$ respectively & $\therefore$ **not Lorentz invariant**

- In QFT, kinetic & potential energy is not talked abt. Instead, **kinetic** and **interacting** terms are used

- All **kinetic terms** are **bilinear**

$$\mathcal{L}_K \supset \frac{1}{2} \phi \square \phi, \bar{\psi} \not{\partial} \psi, \frac{1}{4} F_{\mu\nu}^2, \frac{1}{2} m^2 \phi^2, \frac{1}{2} \phi_1 \square \phi_2, \phi_1 \partial_\mu A_\mu, \dots$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

- Convention dictates use of ϕ, π for scalars, ψ, ξ, χ for fermions, $A_\mu, \bar{U}_\mu, \bar{V}_\mu$ for vectors, $h_{\mu\nu}, T_{\mu\nu}$ for tensors

- Kinetic terms dictate **free** behavior & fields with them are considered **propagating**

- If it has time der. in kinetic it is considered dynamical

- Sometimes it is more convenient to think of **mass** term like $m^2 \phi^2$ as opposed to kinetic
- **Interaction** terms have ≥ 3 fields

$$L_{int} \supset \lambda \phi^3, g \bar{\psi} A \psi, g \partial_\mu \phi A_\mu^\dagger, g^2 f A_\mu^2, \\ \frac{1}{M_{Pl}} \partial_\mu h_{\mu\nu} \partial_\nu h_{\alpha\beta} h_{\alpha\beta}, \dots$$

$$- L_{int} = - \mathcal{V} = - \text{Hint}$$

3.2) Euler Lagrange Equations

- QFT always uses Lagrangians due to L.I.
- Lagrangian system evolution/dynamics determined by **principle of least action** (similar to **Lagrangian classical mech.**)

Action:

$$S = \int dt L = \int d^4x \mathcal{L}$$

Theorem (Euler Lagrange Equations):

- In order to minimize action of $L[d, \partial d]$, the following has to be true:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0$$

Proof:
suppose we have a variation of $\phi \rightarrow \phi + \delta\phi$

Then:

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta(\partial_\mu \phi) \right]$$

$$= \int d^4x \left\{ \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right] \delta\phi + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta\phi \right] \right\}$$

- Last term $(\partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta\phi \right])$ only depends on field values at infinity $\rightarrow$ which we assume the field vanishes at such values

- Safe to ignore

- Allows us to integrate by parts
"without consequences"

$$(\partial_\mu (A B)) = (\partial_\mu A) B + A (\partial_\mu B)$$

$$\text{but } \partial_\mu (A B) = 0$$

$$\text{so } A (\partial_\mu B) = - (\partial_\mu A) B$$

only within Lagrangian

Note:

This is because we integrate over $\mathbb{R}^{3,1}$

so we can use 4 Gauss

- confirming w/ flol proof, we wait

$$\frac{\delta S}{\delta \phi} = 0$$

- we then obtain Euler Lagrange
Equation:

$$\left| \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0 \right|$$

Example:

$$S = \int d^4x \left[\frac{1}{2} (\partial_\mu \phi)(\partial_\mu \phi) - \mathcal{V}(\phi) \right]$$

$$\Leftrightarrow -\mathcal{V}'[\phi] - \partial_\mu (\partial_\mu \phi) = 0$$

$$\text{or } \square \phi + \mathcal{V}'[\phi] = 0$$

$$\text{If } \mathcal{L} = \frac{1}{2} (\partial_\mu \phi)(\partial_\mu \phi) - \frac{1}{2} m^2 \phi^2,$$

we get Klein-Gordon eqn:

$$(\square + m^2) \phi = 0$$

- may restrict to $\mathcal{L}[\phi, \partial\mu\phi]$
- Classically, all Lagrangians are restricted to form $\mathcal{L}[\phi, \partial\mu\phi]$
 - Boundary conditions specified by initial position & velocity
- Kinetic terms w/ too many derivatives like $\partial^2\phi$ are considered "branched"
 - Negative energy states
 - Negative norm
 - still may occur
 - must occur in renormalizable field theories due to quantum effects (such as effective field theories)

3.3 Noether's Theorem

Lagrangians can be invariant under transformations

Example:

$$\mathcal{L} = |\partial_\mu \phi|^2 - m^2 |\phi|^2$$

under $\phi \rightarrow e^{-i\alpha} \phi$ if $\phi \in \mathcal{G}$

Another way to look at this is

$$\phi = \phi_1 + i\phi_2 \text{ by } |\phi|^2 = \phi \phi^*$$

we can obtain

$$\mathcal{L} = (\partial_\mu \phi)(\partial^\mu \phi^*) - m^2 \phi \phi^*$$

and symmetric transformations

are $\phi \rightarrow e^{-i\alpha} \phi$, $\phi^* \rightarrow e^{i\alpha} \phi^*$

(and the equations of motion

are $(\square + m^2)\phi = 0$ & $(\square + m^2)\phi^* = 0$)

- when there is a symmetry parametrized by some α , it is said to be continuous (under the condition that α can be arbitrarily small)

- Similarly as to derivation of E.L. equations, one can find that

$$0 = \frac{\delta \mathcal{L}}{\delta \alpha} = \sum_n \left\{ \left[\frac{\partial \mathcal{L}}{\partial \phi_n} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \right] \frac{\delta \phi_n}{\delta \alpha} + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} \right] \right\}$$

↑
bc symmetry

- When equations of motion (Euler-Lagrange equations) are satisfied,

we obtain

$$\partial_\mu T^\mu = 0$$

where

$$T_\mu \equiv \sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha}$$

Example: (from earlier)

$$\frac{\delta \phi}{\delta \alpha} = -i\phi, \quad \frac{\delta \phi^*}{\delta \alpha} = i\phi^*$$

$$T_\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} (-i\phi) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^*)} (i\phi^*)$$

$$= -i(\phi \partial_\mu \phi^* - \phi^* \partial_\mu \phi)$$

$$\text{Then } \partial_\mu T^\mu = -i(\phi \square \phi^* - \phi^* \square \phi)$$

Since $\square \cdot m. (\square \phi = -m^2 \phi,$
 satisfied, $\square \phi = -m^2 \phi$)

$$\partial_\mu J^\mu = 0$$

- A ^{vector} field satisfying $\partial_\mu J^\mu = 0$ is called a **conserved current**

Noether's theorem:

If a lagrangian has a cont. symmetry, there exists a conserved current associated with that symmetry (when equations of motion satisfied)

proof:

total charge $Q \equiv \int d^3x J_0$

satisfies

$$\partial_t Q = \int d^3x \partial_t J_0 \quad \text{because } \partial_\mu J^\mu = 0$$

$$= \int d^3x \vec{\nabla} \cdot \vec{J} = 0$$

Notes:

- Does not apply to discrete symmetries

- Must be conserved cu-shell (EOM satisfied)
- works for **global** symmetries (parametrized by a) not just local (parametrized by $a(x)$)

Energy Momentum Tensor

- Application of Noether's theorem to global space time translations
- (conserved current is $T_{\mu\nu}$)

- **space time formulation:**
 - Law of physics at x should be same at y (or at $y^\mu = x^\mu + \xi^\mu$) w/ ξ some const. 4-vector

Transformation:

$$\begin{aligned} \phi(x) &\rightarrow \phi(x + \xi) \\ &= \phi(x) + \xi^\mu \partial_\mu \phi(x) + \dots \end{aligned}$$

Transformation Law:

$$\frac{\delta \phi}{\delta \xi^\mu} = \partial_\mu \phi$$

- Applies everywhere (any field)
- Applies to L

$$\frac{\delta L}{\delta g^{\mu\nu}} = \partial_\mu L$$

(This is because $\delta S = \int d^4x \delta L$
 $= \int d^4x \partial_\mu L = 0$)

Derivation of $T_{\mu\nu}$:

$$\frac{\delta L[\phi_\mu, \partial_\mu \phi_\mu]}{\delta g^{\mu\nu}} = \partial_\nu \left(\sum_\mu \frac{\partial L}{\partial (\partial_\mu \phi_\mu)} \frac{\delta \phi_\mu}{\delta g^{\mu\nu}} \right)$$

$$\partial_\nu L = \partial_\mu \left(\sum_n \frac{\partial L}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta g^{\mu\nu}} \right)$$

$$\Rightarrow \partial_\mu \left(\sum_n \frac{\partial L}{\partial (\partial_\mu \phi_n)} \right) \partial_\nu \phi_n - g_{\mu\nu} L = 0$$

$$\boxed{T_{\mu\nu} = \sum_n \frac{\partial L}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} L}$$

gates $\partial_\mu T_{\mu\nu} = 0$

- Energy density,

$$\mathcal{E} = T_{00} = \sum_n \frac{\partial \mathcal{L}}{\partial \dot{\phi}_n} \dot{\phi}_n - \mathcal{L}$$

w/ $\dot{\phi}_n = \partial_4 \phi_n$ obviously

Lagrangian idea with $\phi = \pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$,

$\mathcal{E} = \mathcal{H}$ due to Legendre transform

Energy Momentum Tensor:

$$Q_{\mu\nu} = \int d^3x T_{\mu\nu} \quad T_{\mu\nu} T^{\mu\nu} = 0$$

- $\partial_\mu Q_{\nu} = 0$ so components time independent

- Components are energy (0) and momentum (1, 2, 3)

more info on currents:

- can be Noether currents
- can be external such as moving electron in wire
- can be sources for fields
 - never are fields w/ kinetic terms
- can be placeholders to Lagrangians

$$L = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \phi^* \square \phi - ie A_\mu (\phi^* \partial^\mu \phi - \partial^\mu \phi^* \phi)$$

$$\text{or } \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \phi^* \square \phi - A_\mu J^\mu$$

Coulomb's Law

$$J_\mu(x) : \begin{cases} J_0(x) = \rho(x) = e \delta^3_{\mathbf{x}} \\ \mathbf{J}(x) = 0 \end{cases}$$

$$L = -\frac{1}{4} F_{\mu\nu}^2 - A_\mu J^\mu$$

$$L = -\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu)^2 - A_\mu J^\mu$$

$$= -\frac{1}{2} (\partial_\mu A_\nu)^2 + \frac{1}{2} (\partial_\mu A_\mu)^2 - A_\mu J^\mu$$

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$$\begin{aligned}
\partial_\mu \left(\frac{\partial(\partial_\alpha A_\alpha)}{\partial(\partial_\mu A_\nu)} \right) &= \partial_\mu \left[2(\partial_\alpha A_\alpha) \frac{\partial(\partial_\beta A_\beta)}{\partial(\partial_\mu A_\nu)} \right] \\
&= \partial_\mu \left[2(\partial_\alpha A_\alpha) \frac{\partial(\partial_\beta A_\nu)}{\partial(\partial_\mu A_\nu)} \right] \\
&= \partial_\mu \left[2(\partial_\alpha A_\alpha) \frac{\partial(\partial_\beta A_\nu)}{\partial(\partial_\mu A_\nu)} g'_{\beta\gamma} \right] \\
&= \partial_\mu \left[2(\partial_\alpha A_\alpha) g'_{\beta\mu} g_{\gamma\nu} g_{\beta\gamma} \right] \\
&= \partial_\nu [2\partial_\alpha A_\alpha] \\
&= \boxed{2\partial_\nu(\partial_\alpha A_\alpha)}
\end{aligned}$$

Applying Euler Lagrange equations, one can derive
(with A_μ as our field)

$$-\mathcal{J}_\nu - \partial_\mu (-\partial_\mu A_\nu) - \partial_\nu (\partial_\mu A_\mu) = 0$$

providing

$$\partial_\mu F_{\mu\nu} = \mathcal{J}_\nu$$

which are Maxwell's equations.

Expanding, $\mathcal{J}_\nu = \partial_\mu (\partial_\mu A_\nu - \partial_\nu A_\mu) = \square A_\nu - \partial_\nu (\partial_\mu A_\mu)$

If the Lorenz Gauge ($\partial_\mu A_\mu = 0$)
is chosen for fixing the p.o.f.,
one can find: $\mathcal{J}_\nu = \square A_\nu$,

gives $g_0(\vec{k})$:

$$\boxed{A_2 = \frac{1}{\epsilon} J_2} \quad \left(\frac{1}{\epsilon} \text{ inv. of } \epsilon \right)$$

propagator:

- Init. A_2 def'd by J_2 &
propagates via $\Pi_A = \frac{1}{\epsilon}$

Potential charge e : only electric field
so no mag. pot.

$$A_i = 0$$

$$A_0(x) = \frac{e}{\epsilon} \delta^3(x)$$

2.1.1 Fourier Transform

- Fourier transform of δ -function

$$\text{eg } \int \delta(\vec{k}) d^3k = 1$$

$$\Rightarrow \boxed{\delta^3(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k} \cdot \vec{x}}}$$

Laplacian $\Delta = \partial_{\vec{x}}^2 / \text{lower}$:

$$\Delta^n \delta^3(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} \Delta^n e^{i\vec{k} \cdot \vec{x}} = \int \frac{d^3k}{(2\pi)^3} (-\vec{k}^2)^n e^{i\vec{k} \cdot \vec{x}}$$

$$\text{identify: } \widetilde{[\Delta^n \delta]}(\vec{k}) = (-\vec{k}^2)^n$$

similarly,

$$\square^n \delta^4(x) = \int \frac{d^4k}{(2\pi)^4} \square^n e^{ik_\mu x_\mu} = \int \frac{d^4k}{(2\pi)^4} (-k^2)^n e^{ik_\mu x_\mu}$$

General i def'n:

$$\square^n f(x) = \int \frac{d^4k}{(2\pi)^4} \square^n \tilde{f}(k) e^{ik_\mu x_\mu} = \int \frac{d^4k}{(2\pi)^4} (-k^2)^n \tilde{f}(k) e^{ik_\mu x_\mu}$$

$$\text{so } \widetilde{[\square^n f]}(k) = (-k^2)^n \tilde{f}(k)$$

$\Delta \leftrightarrow -\vec{k}^2$, $\square \leftrightarrow -k^2$ are interchangeable

γ, μ, z More Coulomb

- From earlier

$$A_0(\vec{r}) = \frac{e}{\Delta} g^3(\vec{r}) = -\frac{e}{\Delta} g^3(\vec{r})$$

- Due to *time independence*

Fourier:

$$A_0(\vec{r}) = \int \frac{d^3k}{(2\pi)^3} \frac{e}{k^2} e^{i\vec{k}\cdot\vec{r}}$$

$$= \frac{e}{(2\pi)^3} \int_0^\infty k^2 dk \int d\Omega \int_0^{2\pi} d\phi \frac{1}{k^2} e^{i k r \cos\theta}$$

$$= \frac{e}{(2\pi)^2} \int_0^\infty dk \frac{e^{i k r} - e^{-i k r}}{k}$$

$$= \frac{e}{8\pi^2} \frac{1}{i r} \int_{-\infty}^\infty dk \frac{e^{i k r} - e^{-i k r}}{k}$$

$$= \frac{e}{8\pi^2} \frac{1}{i r} \lim_{\delta \rightarrow 0} \left[\int_{-\infty}^\infty \frac{e^{i k r} - e^{-i k r}}{k + i\delta} \right]$$

$$\text{for } \lim_{\delta \rightarrow 0} \left[\int_{-\infty}^\infty \frac{e^{i k r} - e^{-i k r}}{k + i\delta} \right],$$

pole at $k = -i\delta$

$e^{i k r} \rightarrow 0$ due to having to close curve upwards & so no pole

$e^{-i k r} \Rightarrow$ close down / get:

$$\int_{-\infty}^\infty dk \frac{e^{-i k r}}{k + i\delta} = -(2\pi i) (-e^{-\delta r}) = 2\pi i e^{-\delta r}$$

$$\therefore |A_0(\vec{r})| = \frac{e}{4\pi} \frac{1}{r}$$

$$w/v = |\vec{r}|$$

Green's Functions

have $A_m = \frac{1}{\square} J_m$

- with complicated J_m , can still find A_m with eqn.
- In general, propagators ^{like $\frac{1}{\square}$} very useful - variation, etc.

Grav. Gen:

$$\mathcal{L} = -\frac{1}{2} h \square h + \frac{1}{3} \lambda h^3 + J h$$

EOM: $\square h - \lambda h^2 - J = 0$

$\lambda = 0 \Rightarrow h_0 = \frac{1}{\square} J$

plug $h = h_0 + h_1$

$$\square(h_0 + h_1) - \lambda(h_0 + h_1)^2 - J = 0$$

$$h_1 = O(\lambda)$$

$$\Rightarrow \square h_1 = \lambda h_0^2 + O(\lambda^2)$$

$$h_1 = \frac{1}{\square} J + \lambda \frac{1}{\square} \left(\left(\frac{1}{\square} J \right) \left(\frac{1}{\square} J \right) \right) + O(\lambda^2)$$

gives expansion for h .

Green's function method:

$$\pi \equiv -\frac{1}{\square} / \text{Green's function (operator)}$$

Free propagator field propagation

$$\pi \equiv -\frac{1}{\square} \text{ so that } \square_x \pi(x, y) = -\delta^4(x-y),$$

$$\square_x \equiv g^{\mu\nu} \partial_\mu \partial_\nu$$

explicitly:

$$\pi(x, y) = \int \frac{d^4 k}{(2\pi)^4} e^{i k(x-y)} \frac{1}{k^2}$$

$$\pi(x, y) = \pi(y, x), \quad \square_y \pi(x-y) = -\delta^4(x-y)$$

can rewrite fields:

$$u(x) = \underbrace{\int d^4 y \delta^4(x-y) u(y)}_{\text{ftd. } \delta\text{-func}} = - \int d^4 y [\square_y \pi(x, y)] u(y)$$

$$= - \int d^4 y \pi(x, y) \square_y u(y)$$

we obtain:

$$u(x) = - \int d^4 y \pi(x, y) \bar{J}(y)$$

$$\square_x u(x) = \lambda u^2(x) \Rightarrow \int d^4 y \pi(x, y) \bar{J}(y) \int d^4 z \pi(x, z) \bar{J}(z)$$

we can obtain via the perturbation:

$$u(x) = - \int d^4 y \pi(x, y) \bar{J}(y)$$

$$+ \lambda \int d^4 x \int d^4 y \int d^4 z \pi(x, y) \pi(x, z) \bar{J}(y) \bar{J}(z) + O(\lambda^2)$$

Can be shown as a picture / Feynman
 diagram:

$$u(x) = \underbrace{}_{J(y)} + \underbrace{}_{J(y)} + \dots$$

Governed by Feynman rules:

Rules for this CF ex.:

1. Draw point x and line from x to x ;
 2. Truncate at source J or split into two lines w/ new point, back of λ ;
 3. repeat
 4. $u(x)$ given by graphs up to some order in λ , ends capped at $J(x_i)$, lines are replaced by propagators $\pi(x_i, x_j)$, integrated over all points
- rules for BKT are almost identical except " $\pi \neq 0$ lines can close in on themselves"
- first diagram for $u(x)$ is Newton's potential.

Problem:

3.1) Derive general E.L. equations / use $L[\phi, \partial_\mu \phi, \partial_2 \partial_\mu \phi, \dots]$

$$S = \int d^4x L[\phi, \partial_\mu \phi, \partial_2 \partial_\mu \phi, \dots]$$

$$\Rightarrow \delta S = \int d^4x \delta L$$

Achieve w.r. $\phi \rightarrow \phi + \delta\phi$

$$\delta S = \int d^4x \left[\frac{\partial L}{\partial \phi} \delta\phi + \frac{\partial L}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) + \frac{\partial L}{\partial(\partial_2 \partial_\mu \phi)} \delta(\partial_2 \partial_\mu \phi) + \dots \right]$$

expand:

$$\begin{aligned} \cancel{\frac{\partial L}{\partial \phi} \delta\phi} + \frac{\partial L}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) &= \left[\frac{\partial L}{\partial \phi} - \partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)} \right] \delta\phi + \partial_\mu \left[\frac{\partial L}{\partial(\partial_\mu \phi)} \delta\phi \right] \\ &= \frac{\partial L}{\partial \phi} \delta\phi - \cancel{\partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)} \delta\phi} \\ &\quad + \cancel{\partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)}} + \frac{\partial L}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \end{aligned}$$

note that $\delta(\partial_\mu \phi) = \partial_\mu(\delta\phi)$

Min. var

$$\frac{\partial L}{\partial \phi} \delta\phi + \frac{\partial L}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) + \frac{\partial L}{\partial(\partial_2 \partial_\mu \phi)} \delta(\partial_2 \partial_\mu \phi) + \dots$$

$$\left[\frac{\partial L}{\partial \phi} - \partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)} - \partial_2 \partial_\mu \frac{\partial L}{\partial(\partial_2 \partial_\mu \phi)} + \dots \right] \delta\phi$$

$$+ 2_m \left[\frac{\partial \mathcal{L}}{\partial (2_m \phi)} \delta \phi \right] + 2_2 2_m \left[\frac{\partial \mathcal{L}}{\partial (2_2 2_m \phi)} \delta \phi \right] + \dots$$

by product rule/int. by parts.

Note that all terms except first (with $\delta \phi$ as common factor) are total derivatives & drop off at infinites

$$\text{want } \delta^2 \mathcal{L} = 0 / \text{have:}$$

$$\left(\frac{\partial \mathcal{L}}{\partial \phi} - 2_m \frac{\partial \mathcal{L}}{\partial (2_m \phi)} - 2_2 2_m \frac{\partial \mathcal{L}}{\partial (2_2 2_m \phi)} + \dots = 0 \right)$$

2.3)

$$a) \mathcal{L} \rightarrow \mathcal{L} + 2_m X^{\mu}$$

$$\mathcal{T}_{\mu\nu} = \left\langle \frac{\partial \mathcal{L}}{\partial (2_m \phi_\mu)} \right\rangle 2_\nu \phi_\nu - g_{\mu\nu} \mathcal{L}$$

$$\rightarrow \left\langle \left[\frac{\partial \mathcal{L}}{\partial (2_m \phi_\mu)} + \frac{\partial (2_m X^\mu)}{\partial (2_m \phi_\mu)} \right] 2_\nu \phi_\nu - g_{\mu\nu} \mathcal{L} - g_{\mu\nu} 2_m X^\mu \right\rangle$$

$$= \left\langle \frac{\partial \mathcal{L}}{\partial (2_m \phi_\mu)} \right\rangle 2_\nu \phi_\nu - g_{\mu\nu} \mathcal{L} + \cancel{\left\langle \frac{\partial (2_m X^\mu)}{\partial (2_m \phi_\mu)} \right\rangle 2_\nu \phi_\nu} - g_{\mu\nu} 2_m X^\mu$$

$$\frac{\partial f}{\partial (2\pi\phi)} g(2\pi\phi)$$

$$2\pi \left(\frac{\partial f}{\partial (2\pi\phi)} \right) g\phi$$

$$= 2\pi \frac{\partial f}{\partial (2\pi\phi)} g\phi + \frac{\partial f}{\partial (2\pi\phi)} 2\pi(g\phi)$$

$$\oint_{\gamma} \omega \wedge (u^2 - \alpha^2)$$

$$\omega = u^n$$

$$du = 2u du$$

$$\rightarrow \oint \frac{du}{2u}$$